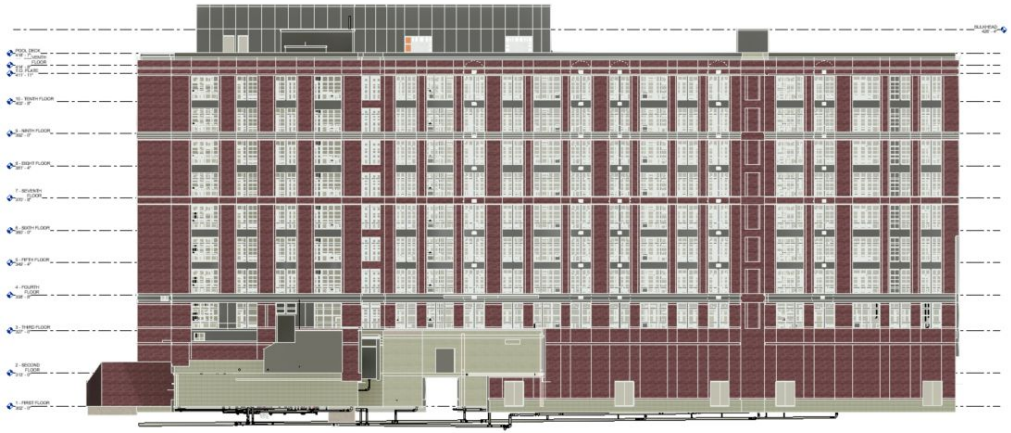
Diego Rueda.

Plumbing & Mechanical Piping
BIM Specialist

Portfolio







DIEGO RUEDA.

Plumbing & Mechanical Piping BIM Specialist | Revit & Fabrication Parts | Navisworks Coordination

EDUCATION

Universidad Industrial de Santander
2013 - 2020
Mechanical Engineering

ABOUT

Mechanical Engineer with over 3 years of BIM experience on plumbing and mechanical piping for U.S. commercial, residential and institutional projects. I build **construction-ready Revit MEP models with Autodesk Fabrication Parts**, detailed for fabrication and field installation, not only design intent.

I run **multidisciplinary coordination and clash detection in Navisworks** with architectural, structural, HVAC, electrical and fire protection teams, and join weekly coordination and constructability meetings with U.S. engineers, general contractors and trade partners. My work includes shop drawings, sleeve drawings and model QC.

ADDITIONAL COURSES

Automated Technology for Building Water Management.
American Society of Plumbing Engineers (ASPE), November 2025
Intro to Heat Pump Water Heaters.
American Society of Plumbing Engineers (ASPE), October 2025
Design of Domestic Hot Water System Part 1.
American Society of Plumbing Engineers (ASPE), July 2025
Design of Domestic Hot Water System Part 2.
American Society of Plumbing Engineers (ASPE), July 2025
Pipe Hangers and Supports.
American Society of Plumbing Engineers (ASPE), July 2025
Piping Insulation.
American Society of Plumbing Engineers (ASPE), July 2025

SOFTWARE

Revit	●●●●●
Navisworks	●●●●○
AutoCAD	●●●○○
ReCap	●●○○○
CADmep	●●○○○
Bluebeam	●●○○○
Blender	●●○○○
ACC/Forma	●●○○○

Others: Fabrication Parts, Python, Dynamo, pyRevit

LANGUAGES

Spanish	●●●●●
English	●●●●○
French	●○○○○

ENG
BIM Specialist (Plumbing and Mechanical Piping)
Jul 2026 - Present

Main task list:
• Model plumbing and mechanical piping in Revit with **Autodesk Fabrication Parts**, ready for fabrication and installation.
• Run **Navisworks clash detection** with architectural, structural, HVAC, electrical and fire protection teams.
• Review installation sequencing with general contractors and trade partners in constructability meetings.

Utopia Design
Plumbing Design Engineer
Apr 2025 - Jun 2026

Main task list:
• Built and maintained **Revit MEP models** and coordinated construction documents for luxury hotels in several countries.
• **Coordinated in Navisworks** with architecture, structure, HVAC and electrical, resolving clashes and optimizing layouts.
• Designed water, drainage, rainwater and greywater systems, including spas and pools.

M&S Mechanical LLC
Plumbing Drafting Lead
Jan 2025 - Sep 2025

Main task list:
• Led the team building **complex Revit MEP plumbing models** for U.S. commercial, residential and institutional projects.
• **Coordinated models in Navisworks** and reviewed shop drawings for accuracy and constructability.
• Mentored plumbing drafters and promoted BIM best practices across deliverables.

M&S Mechanical LLC
MEP Engineer
Sep 2023 - Dec 2024

Main task list:
• **Modeled domestic water, sanitary, vent and storm** systems in Revit MEP and CADmep.
• Produced coordinated **shop drawings for fabrication** and field installation.
• Ran clash detection and model QC in Navisworks and checked designs against U.S. plumbing codes.

US-Based Projects

Five U.S. projects modeled as BIM Specialist – 3D Modeler

Revit Fabrication Parts on every project · Navisworks clash tests run on my end to speed up coordination · Sleeve and shop drawings

#	PROJECT	SCOPE	SYSTEMS & DELIVERABLES
01	**Multifamily Residential Building** Residential	Mid-rise residential building with retail, pool and amenities	Sanitary, storm, vent, domestic cold and hot water with recirculation; sleeve drawings per floor
02	**Parking Garage** Parking	Multi-level parking structure	Sanitary, storm, vent, domestic cold and hot water with recirculation; sleeve and shop drawings per floor
03	**Residential Tower** Residential	High-rise residential tower with amenities and common areas	Sanitary, storm, vent, domestic cold and hot water with recirculation; sleeve and shop drawings per floor
04	**Temple** Religious	Multi-level temple with utility yard	Storm, sanitary and vent, domestic water, natural gas and baptismal font
05	**Meditation Center** Religious	Single-level meditation center	Domestic water with water heaters, sanitary and vent, storm, sewage ejector and sump

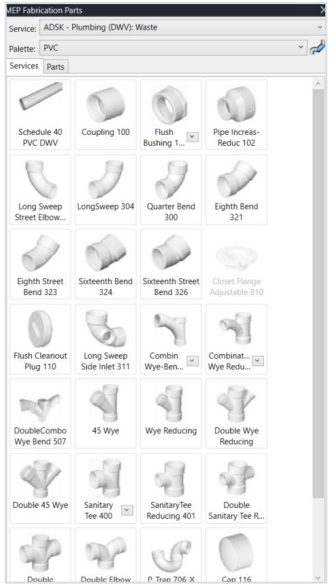
Fabrication Parts Databases

Materials: Multiple pipe materials; PVC, cast iron and copper shown
Software: Revit with Autodesk Fabrication Parts
Used on: Every project that follows

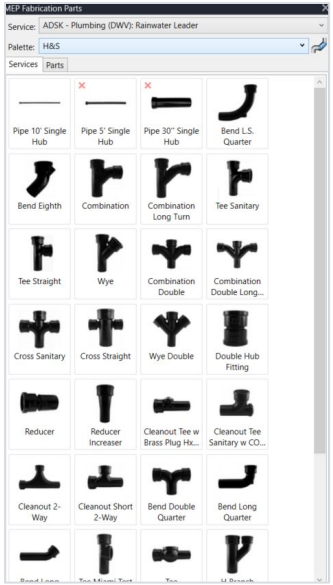
Fabrication-level models: every project in this portfolio was modeled with Fabrication Parts databases instead of generic Revit pipe types.

Real parts: pipes and fittings carry each database's material, sizes and connection types, so the model reflects what gets installed rather than design intent.

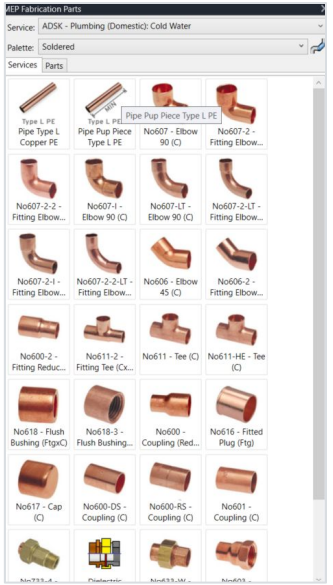
Coordination: the same fabrication models are checked for clashes in Navisworks, as shown on the Temple slides.



PVC database



Cast iron database



Copper database

US-Based Project

Multifamily Residential Building

Type of Project: Mid-rise multifamily residential with retail and amenities
Location: United States
Deliverable: Plumbing construction documents

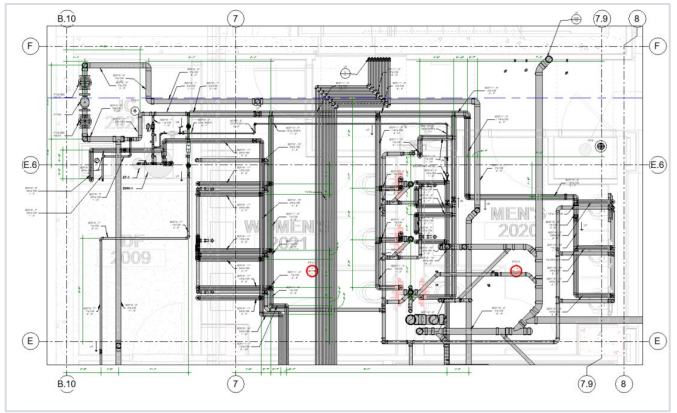
Scope: Residential building with ground-floor retail, pool and common areas.

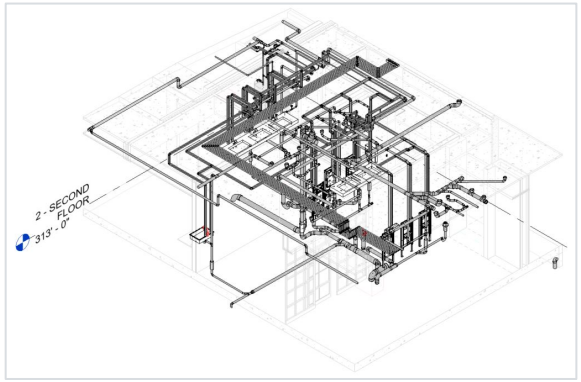
Systems: Sanitary, vent, storm, domestic cold and hot water with recirculation, booster and sump pumps, natural gas distribution.

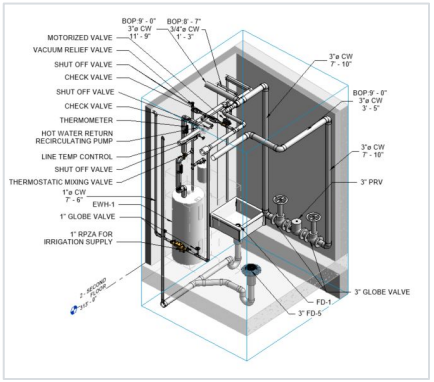
Drawing Set: Plumbing floor plans by level, gas distribution plans and plumbing schedules.

My Work: Fabrication-level plumbing model in Revit with Autodesk Fabrication Parts databases; Navisworks clash tests run on my end to speed up coordination.

Role: BIM Specialist – 3D Modeler
Location: Remote







Bathroom detail.
3D detail views and an enlarged plan with dimensions and pipe tags, issued for field installation.

US-Based Project

Multifamily Residential Building: Sleeve Layout & Takeoff

Sleeve layout for Level 4, Section B, and quantity takeoffs of square and round sleeves for the whole of Level 4.



Sleeve layout. Level 4, Section B.

<SLEEVES SCHEDULE 4TH FLOOR>

A	B	C
Type	LENGTH	Count
3"ø Sleeve	0' - 3 1/2"	119
4"ø Sleeve	0' - 3 1/2"	98
5"ø Sleeve	0' - 3 1/2"	131
6"ø Sleeve	0' - 3 1/2"	80
8"ø Sleeve	0' - 3 1/2"	10
Grand total: 438		

Square sleeves takeoff. Whole of Level 4.

<BOX SCHEDULE 4TH FLOOR>

A	B	C	D	E
Type	Width	Height	Length	Count
BOX	0' - 3"	0' - 9"	0' - 3 1/2"	1
BOX	0' - 3"	1' - 4"	0' - 3 1/2"	20
BOX	0' - 3"	1' - 6"	0' - 3 1/2"	1
BOX	0' - 3"	1' - 8"	0' - 3 1/2"	1
BOX	0' - 5"	1' - 0"	0' - 3 1/2"	12
BOX	0' - 6"	1' - 6"	0' - 3 1/2"	2
BOX	0' - 8"	1' - 4"	0' - 3 1/2"	2
BOX	0' - 10"	0' - 4"	0' - 2 1/2"	23
Total: 62				

Round sleeves takeoff. Whole of Level 4.

US-Based Project

Temple

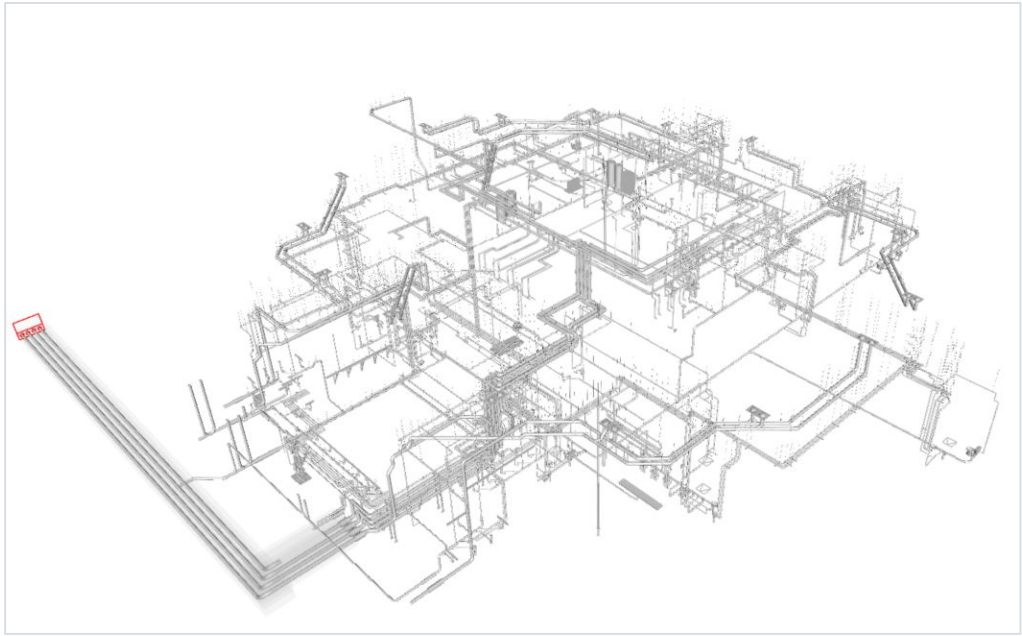
Type of Project: Religious building, multiple levels
Location: United States
Deliverable: Plumbing construction documents

Scope: Temple with several levels plus a utility yard.

Systems: Storm, sanitary and vent, domestic water with booster and backflow, natural gas and baptismal font.

Drawing Set: Plans by level and area, sanitary/vent and domestic isometrics, details, schematics, schedules and gas diagrams.

My Work: Coordination model of **DWV, domestic water and mechanical piping with hangers**, built in Revit with **Autodesk Fabrication Parts databases**; Navisworks clash tests run on my end to resolve conflicts before coordination.

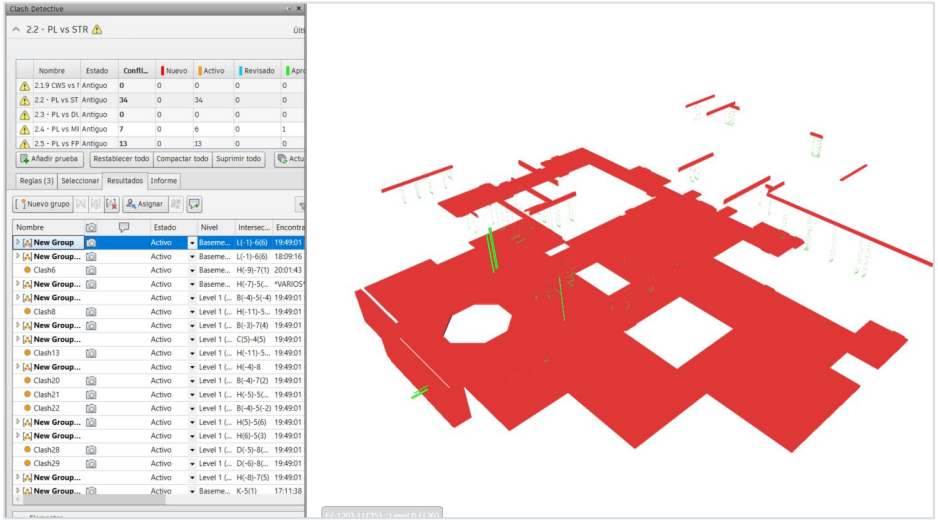


Coordination model. DWV, domestic water and mechanical piping, modeled with hangers for installation.

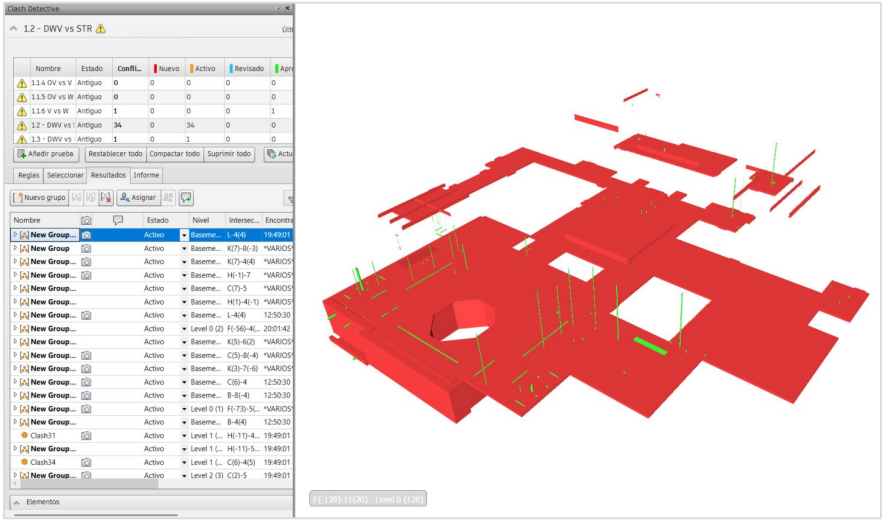
Role: BIM Specialist – 3D Modeler
Location: Remote

Temple: Navisworks Clash Detection

Clash detection tests I set up in Navisworks to find conflicts between the plumbing model and the other trades before coordination meetings.



Clash test 1: PL vs. STR. Plumbing water piping checked against the structural model.



Clash test 2: DWV vs. STR. Drain, waste and vent piping checked against the structural model.

US-Based Project

Meditation Center

Type of Project: Religious / assembly building
Location: United States
Deliverable: Plumbing construction documents

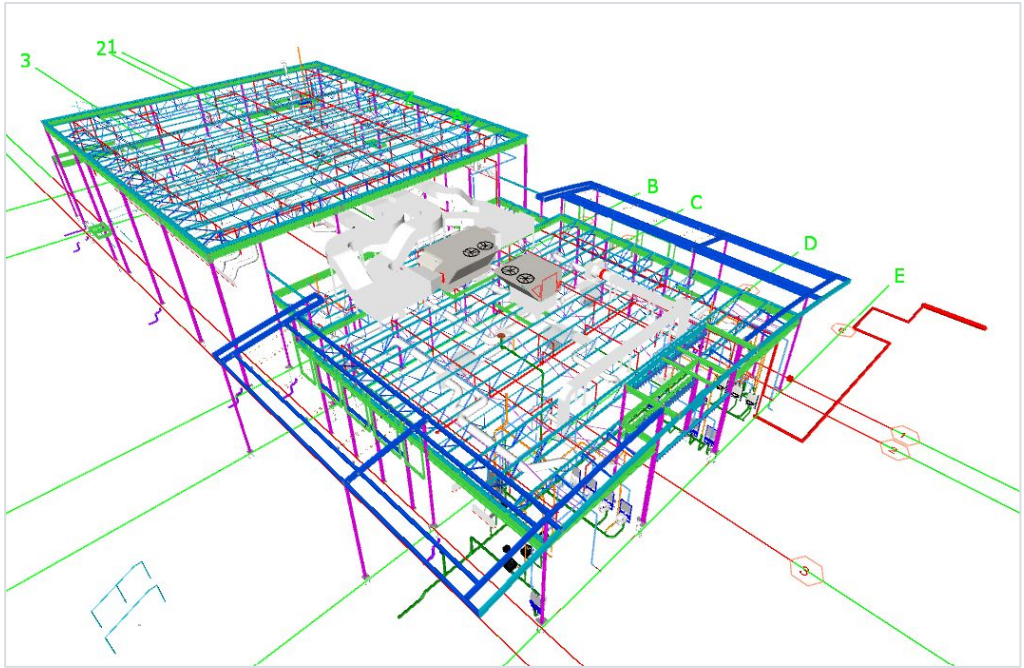
Scope: Single-level meditation center with underground, ground-floor and roof plumbing.

Systems: Domestic cold and hot water with water heaters, sanitary waste and vent, storm drainage, sewage ejector and sump.

Drawing Set: Site, underground, floor and roof plans, enlarged plans, riser diagrams, details, equipment schedule and energy compliance forms.

My Work: Fabrication-level plumbing model in Revit with Autodesk Fabrication Parts databases; Navisworks clash tests run on my end to speed up coordination.

Role: BIM Specialist – 3D Modeler
Location: Remote



Federated coordination model. Plumbing model coordinated with all trades involved in the project, used for Navisworks clash detection.

Thank you.

Plumbing & Mechanical Piping
BIM Specialist

Contact

EMAIL

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LINKEDIN

linkedin.com/in/diegoarueda

LOCATION

Remote for U.S. projects